A3.1 Diversity of organisms

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a3-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 102–117 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

WHAT IS A SPECIES?

A3.1.1 Variation is a defining feature of life

pp. 102–103

No two individuals are identical in all their traits. These complex patterns of variation are the basis for naming and classifying organisms.

A3.1.2 Species as groups with shared traits

pp. 103–104

Linnaeus grouped organisms by what they looked like and how they were built. This is useful, but it can mislead: some species look alike, and some members of one species look quite different.

A3.1.3 Binomial names

pp. 104–106

A two-part name gives the **genus**, with a capital letter, and then the **species**, in lower case. Names are printed in italics and underlined when handwritten, as in *Homo sapiens*. Species in one genus have similar traits, and the names are understood worldwide.

A3.1.4 The biological species concept

pp. 106–107

A **species** is a group that can breed and produce **fertile offspring**. A horse and a zebra can produce young, but the young are normally infertile. The concept has limits: it cannot be used for asexual organisms or fossils, and other species concepts exist.

A3.1.5 Populations, species and speciation

pp. 107–109

Speciation is one species splitting into two or more. It is usually gradual, with non-interbreeding populations becoming more different over time. There is no single point at which two populations clearly become two species.

CHROMOSOMES

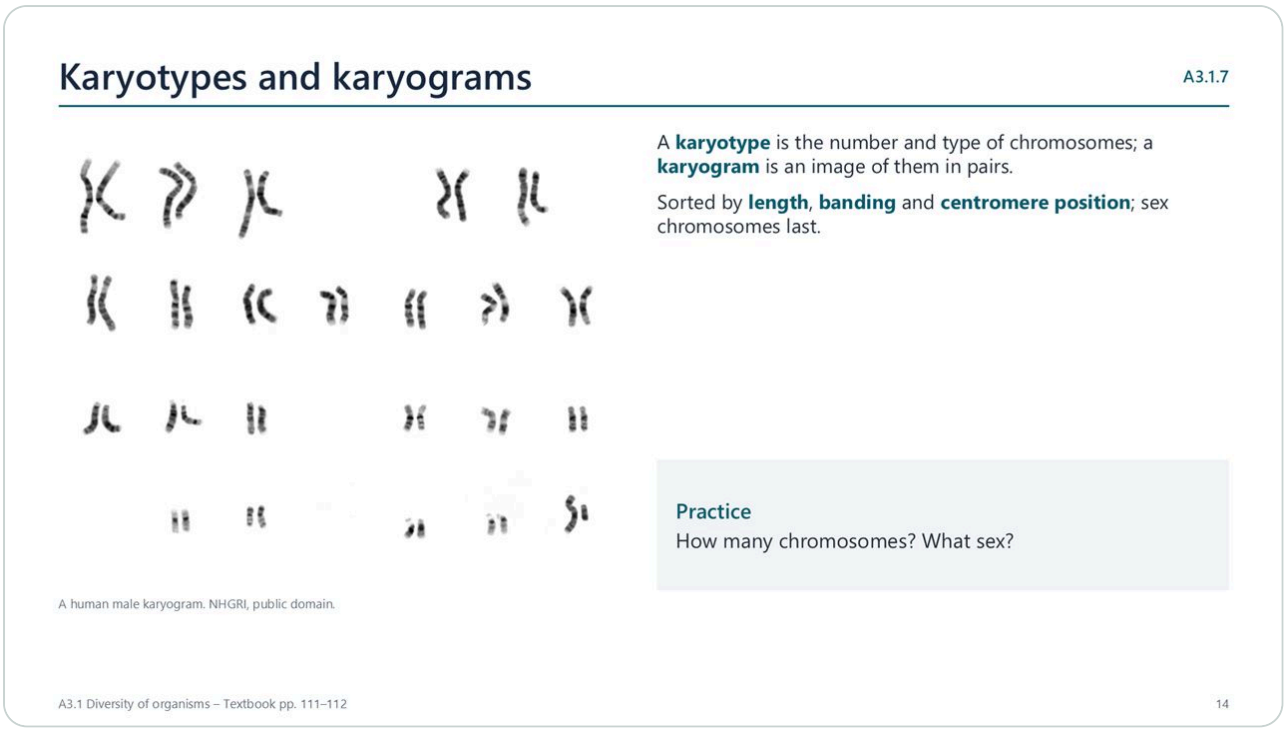
A3.1.6 Chromosome numbers vary

p. 110

Humans have 46 chromosomes and chimpanzees have 48. Numbers vary widely between plants and animals, and they do not show how complex an organism is.

A3.1.7 Karyotypes and karyograms

pp. 111–113



A **karyogram** shows chromosomes in pairs, sorted by **length**, **banding pattern** and **centromere position**. Sex chromosomes come last.

Human chromosome 2 seems to have formed by the **fusion** of two ancestral chromosomes. It has telomere sequences near its middle, a second inactive centromere, and bands matching two chimpanzee chromosomes.

Watch out: humans did not evolve from chimpanzees. Both evolved from a shared ancestor.

GENOMES

A3.1.8 Genomes within a species

p. 115

A **genome** is all of an organism's genetic information. Members of a species share most of their genome, but variations such as **single-nucleotide polymorphisms** give some diversity.

A3.1.9 Diversity of eukaryotic genomes

pp. 115–116

Genomes vary in total size and in base sequence. Differences between species are much larger than differences within a species.

A3.1.10 Comparing genome sizes

p. 116

Data from genome databases show that size does not simply follow complexity. A lungfish has a genome many times larger than a human's, because much DNA does not code for proteins.

A3.1.11 Current and future uses of sequencing

p. 117

Sequencing has become far faster and cheaper. It is now used to research **evolutionary relationships**, and may in future allow **personalized medicine**.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission